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01 Buchholz's Ordinal Collapsing Function

Buchholz's **Ordinal Collapsing Function** (BOCF) is a way for us to create even larger (countable) ordinals. Ironically, to make these larger ordinals, we will be using even larger *uncountable* ordinals to get there.

But first, how exactly are ordinals of higher cardinality ($\omega_1, \omega_2, \ldots$) defined? Let's first define the *successor cardinal* of $\aleph_0$, $\aleph_1$, and its assigned ordinal ω_1 as **the set of all ordinals whose cardinality $\leq \aleph_0$**:

$$\aleph_1 = \omega_1 = \{\alpha \in \mathrm{Ord} : |\alpha| \leq \aleph_0\}$$

where $|\alpha|$ represents the cardinality of an ordinal α. Similarly, we have the cardinal $\aleph_2$ and its assigned ordinal ω_2 to be the set of all ordinals whose cardinality is $\leq \aleph_1$. We can repeat this to make $\aleph_3, \aleph_4, \ldots$. For a limit ordinal λ, we define $\aleph_\lambda$ as such:

$$\aleph_\lambda = \bigcup_{\beta<\lambda} \aleph_\beta$$

Definition 1 (**Buchholz's ψ function**)

Let Ω_ν be defined as such:

$$\Omega_\nu := \begin{cases} 1 & \text{if } \nu = 0 \\ \omega_\nu & \text{if } \nu > 0 \end{cases}$$

Buchholz's ordinal collapsing function ψ and the set $C_\nu(\alpha)$ is defined as such:

1. $\Omega_\nu \subseteq C_\nu(\alpha)$
2. For any two ordinals $\xi, \eta \in C_\nu(\alpha)$, their sum $\xi + \eta \in C_\nu(\alpha)$
3. For any ordinal $\xi \in C_\nu(\alpha)$, as long as $\xi < \alpha$, then $\psi_\mu(\xi) \in C_\nu(\alpha)$ for all $\mu \leq \omega$
4. $\psi_\nu(\alpha) = \min\{\gamma \in \mathrm{Ord} \mid \gamma \notin C_\nu(\alpha)\}$, i.e., the smallest ordinal not inside $C_\nu(\alpha)$

This is a lot to unpack, so let's first start with $\nu = 0$:

Example 1

Let's first find what $\psi_0(0)$ is.

In order to figure out what $\psi_0(0)$, we first need to know what's inside $C_0(0)$. Going down the list, we have:

1. $\Omega_0 = 1 = \{0\} \subseteq C_0(0)$

   so currently $C_0(0) = \{0\}$

2. For any two ordinals in $C_0(0)$, their sum is also in $C_0(0)$

   Another way to interpret this statement is to say that you can take a collection of ordinals in $C_0(0)$ and apply addition a finite number of times on them. In this case, all we have is 0, so nothing new gets added here.

3. For any ordinal $\xi \in C_0(0)$, as long as $\xi < 0$, then $\psi_\mu(\xi) \in C_0(0)$ for all $\mu \leq \omega$

In this case, there are no ordinals $\xi < 0$ so, nothing new get added here either.

4. $\psi_0(0)$ is the smallest ordinal *not* contained inside $C_0(0) = \{0\}$, which is just 1.

In fact, we can generalize this and show that:

Lemma 1

$\psi_\nu(0) = \Omega_\nu$

Proof

From the definition, $\Omega_\nu \subseteq C_\nu(0)$, i.e. for all $\alpha \in \Omega_\nu, \alpha \in C_\nu(0)$.

For all $\alpha, \beta \in \Omega_\nu, \alpha + \beta \in \Omega_\nu$, so adding doesn't generate us anything new.

Since there are no ordinals $\xi < 0$, we can't use ψ to generate us anything new.

So $C_\nu(0) = \{\gamma \in \mathrm{Ord} \mid \gamma < \Omega_\nu\}$. Then $\psi_\nu(0)$ is the smallest ordinal not in $C_\nu(0)$ which is Ω_ν.

Example 2

Let's proceed to find the value of $\psi_0(1)$:

1. $\Omega_0 \subseteq C_0(0)$, so we once again have $0 \in C_0(0)$.
2. Adding doesn't do anything

The difference this time is that we can use $\psi_0(0)$:

3. For any ordinal $\xi \in C_0(0)$, as long as $\xi < 1$, then $\psi_{\mu(\xi)} \in C_0(0)$ for all $\mu \leq \omega$

   The only valid values of ξ is just 0, so we have $\psi_0(0), \psi_1(0), \ldots, \psi_{\omega(0)} \in C_0(0)$. $\psi_0(0) = 1 \in C_0(0)$, so we finally have something new. We haven't gotten to $\psi_1(0), \psi_2(0), \ldots, \psi_{\omega(0)}$ yet, and we'll ignore them for now.
2. We can now use addition to show that $1 + 1 = 2 \in C_0(0), 2 + 1 = 3 \in C_0(0)$, and so on to show that all natural numbers $\mathbb{N} \subseteq C_0(0)$. In fact, $\mathbb{N} = C_0(0)$.
4. $\psi_0(1)$ is the smallest ordinal not in $C_0(0)$, which is ω

1 Properties

Before we continue further, let's look at some properties about this function. Let's first introduce the concept of an **additively principal** ordinal:

Definition 2

An **additively principal** ordinal α is any non-zero ordinal such that for any $\beta, \gamma < \alpha$, $\beta + \gamma < \alpha$.

We denote the *class* of all additively principal numbers as P. P is precisely all ordinals of the form ω^ξ, where ξ is some ordinal.

Formally, we can define P as such:

$$\begin{aligned} P &:= \{\alpha \in \mathrm{Ord} \mid 0 < \alpha \text{ and for all } \beta, \gamma < \alpha, \beta + \gamma < \alpha\} \\ &= \left\{\omega^\xi \mid \xi \in \mathrm{Ord}\right\} \end{aligned}$$

We can also define a function $P(\alpha)$ that "decomposes" an ordinal α into its additively principal "components":

Definition 3

We define $P(\alpha)$ as:
1. $P(0) := \emptyset$
2. For $\alpha > 0$, there are uniquely determined $\alpha_0, \alpha_1, ..., \alpha_n \in P$ with $\alpha = \alpha_0 + \alpha_1 + ... + \alpha_n$.

   We set $P(\alpha) := \{\alpha_0, \alpha_1, ..., \alpha_n\}$

This notion is useful as the Buchholz function ψ, as each $C_\nu(\alpha)$ represents all ordinals that can be derived from addition and the ψ function itself:

Lemma 2

1. If $\zeta, \eta \in C_\nu(\alpha)$, then $\zeta + \eta \in C_\nu(\alpha)$
2. An ordinal $\gamma \in C_\nu(\alpha)$ if and only if $P(\gamma) \subseteq C_\nu(\alpha)$
3. If $\zeta + \eta \in C_\nu(\alpha)$, then $\eta \in C_\nu(\alpha)$

Proof

1. In the definition
2. Since $\gamma = \gamma_0 + \gamma_1 + ... + \gamma_n$ where $\gamma_i \in P(\gamma)$, and $P(\gamma) \subseteq C_\nu(\alpha)$, then $\gamma_i \in C_\nu(\alpha)$ for all $i \leq n$, and as such, we can "reconstruct" γ from elements of $C_\nu(\alpha)$:
$$\gamma = \gamma_0 + ... + \gamma_n \in C_\nu(\gamma)$$
3. Since $\xi + \eta \in C_\nu(\alpha)$, we have $P(\eta) \subseteq P(\eta + \xi) \subseteq C_\nu(\alpha)$, and thus, $\eta \in C_\nu(\alpha)$

We can additionally show that the ψ function only outputs additively principal ordinals:

Lemma 3

$\psi_\nu(\alpha) \in P$ for all ordinals ν, α.

Proof

Suppose $\psi_\nu(\alpha) \notin P$. Then $P(\psi_\nu(\alpha)) \subseteq \psi_\nu(\alpha)$, which is to say, for every element $p \in P(\psi_\nu(\alpha))$, $p < \psi_\nu(\alpha)$, which also means $p \in C_\nu(\alpha)$, thus $P(\psi_\nu(\alpha)) \subseteq C_\nu(\alpha)$. However, this implies $\psi_\nu(\alpha) \in C_\nu(\alpha)$, a contradiction.

Another interesting property of this function is that subscript of ψ represents the cardinality of it's result:

Proposition 1

$\Omega_\nu \leq \psi_{\nu(\alpha)} < \Omega_{\nu+1}$

Proof

Since $\Omega_\nu \subseteq C_\nu(\alpha)$, we get $\Omega_\nu \leq \psi_\nu(\alpha)$. Since the cardinality of $C_\nu(\alpha)$ is less than $\Omega_{\nu+1}$, we have $\psi_\nu(\alpha) < \Omega_{\nu+1}$.

So ψ_0 will only spit out countable ordinals $1 \leq \psi_0(\alpha) < \Omega$, ψ_1 will only be in the range $\Omega \leq \psi_1 < \Omega_2$, etc....

2 Bigger ordinals

Now let's look at $\psi_0(2)$. Since $C_0(1) \subseteq C_0(2)$, we will mainly focus on what *additional* elements are contained with $C_0(2)$. For starters, since $1 < 2$, we can access $\psi_0(1) = \omega, \psi_1(1), \ldots$. Since we can also use addition, ordinals like $\omega + 1$, $\omega + \omega$, or even $\omega \cdot 1000$ are within $C_0(2)$. However, as we cannot add ω to itself transfinitely many times, the least ordinal not contained within $C_0(2)$ is $\psi_0(2) = \omega^2$.

So far, as we have increased α, $\psi_0(\alpha)$ has increased. In fact, this is true for any ψ_ν:

Lemma 4

1. If $\alpha \leq \beta$, then $C_\nu(\alpha) \subseteq C_\nu(\beta)$ and $\psi_\nu(\alpha) \leq \psi_\nu(\beta)$
2. If $\alpha < \beta$ and $\alpha \in C_\nu(\alpha)$, then $\psi_\nu(\alpha) < \psi_\nu(\beta)$

Proof

1. Trivial
2. From the previous statement, we can conclude that $C_\nu(\alpha) \subseteq C_\nu(\beta)$ and $\psi_\nu(\alpha) \leq \psi_\nu(\beta)$. Since $C_\nu(\alpha) \subseteq C_\nu(\beta)$ and $\alpha \in C_\nu(\alpha)$, $\alpha \in C_\nu(\beta)$, and since $\alpha < \beta$, $\psi_\nu(\alpha) \in C_\nu(\beta)$

Lemma 5

If $\gamma = \psi_{\nu_1}(\alpha_1) = \psi_{\nu_2}(\alpha_2)$, then $\nu_1 = \nu_2$ and $\alpha_1 = \alpha_2$, i.e., all outputs of ψ are uniquely represented.

Proof

Left as an exercise to the reader

Continuing from where we left off, we can repeat this for $C_0(3)$ to get $\psi_0(3) = \omega^3$, $\psi_0(4) = \omega^4$ and so on. It seems like $\psi_0(\alpha) = \omega^\alpha$, and we can prove that fact: We first prove a property about how $\psi_\nu(\alpha + 1)$ relates to $\psi_\nu(\alpha)$:

Lemma 6

1. $\psi_\nu(\alpha + 1) = \min\{\gamma \in P \mid \psi_\nu(\alpha) < \gamma\}$ if $\alpha \in C_\nu(\alpha)$
2. $\psi_\nu(\alpha + 1) = \psi_\nu(\alpha)$ if $\alpha \notin C_\nu(\alpha)$
3. If α is a limit ordinal, $\psi_\nu(\alpha) = \sup^{+}\{\psi_\nu(\xi) \mid \xi < \alpha \text{ and } \xi \in C_\nu(\xi)\}$

Proof

[WIP]

1. Since $\alpha \in C_\nu(\alpha)$, and $\alpha \in C_\nu(\alpha)$, $\psi_\nu(\alpha) < \psi_\nu(\alpha+1)$. Suppose for contradiction there exists an ordinal γ such that $\psi_\nu(\alpha) \leq \gamma < \psi_\nu(\alpha+1)$, and $\gamma \in P$.

Lemma 7

1. For all ordinals $\alpha < \varepsilon_0$, $\alpha \in C_0(\alpha)$ and $\psi_0(\alpha) = \omega^\alpha$
2. For all ordinals $\alpha < \varepsilon_{\Omega_\nu+1}$ where $\nu \neq 0$, $\alpha \in C_\nu(\alpha)$ and $\psi_\nu(\alpha) = \omega^{\Omega_\nu+\alpha}$

Proof

Suppose $\alpha \in C_0(\alpha)$ and $\psi_0(\alpha) = \omega^\alpha$. Then $\alpha+1 \in C_0(\alpha+1)$ and $\psi_0(\alpha+1)$

Using the lemma above, we have:

$$\begin{aligned}
\psi_0(1) &= \omega \\
\psi_0(2) &= \omega^2 \\
\psi_0(\omega) &= \omega^\omega \\
\psi_0(\omega+1) &= \omega^{\omega+1} \\
\psi_0(\omega^\omega) &= \omega^{\omega^\omega}
\end{aligned}$$

Then what about $\psi_0(\varepsilon_0)$?

$$\begin{aligned}
\psi_0(\varepsilon_0) &= \overset{+}{\sup}\{\psi_0(\xi) \mid \xi < \varepsilon_0 \text{ and } \xi \in C_0(\xi)\} \\
&= \left\{\omega^\xi \mid \xi < \varepsilon_0\right\} \\
&= \varepsilon_0
\end{aligned}$$

Therefore we have $\psi_0(\varepsilon_0) = \varepsilon_0$, and $\varepsilon_0 \notin C_0(\varepsilon_0)$. As such, we can't use ε_0 in $C_0(\varepsilon_0+1)$ (i.e., $\psi_0(\varepsilon_0) = \varepsilon_0 \notin C_0(\varepsilon_0+1)$), and $C_0(\varepsilon_0+1) = C_0(\varepsilon_0+1)$. So $\psi_0(\varepsilon_0+1) = \psi(\varepsilon_0) = \varepsilon_0$ In fact, none of the ordinals we have mentioned will cut it, $\psi_0(\Gamma_0) = \psi_0(\text{SVO}) = \psi_0(\text{LVO}) = \varepsilon_0$. This is where $\psi_1()$ comes in.

We (now) know that $\psi_1(0) = \Omega_1$. If you recall from the start, our $C_0(1)$ contains not only $\psi_0(0) = 1$, but also $\psi_1(0) = \Omega_1, \psi_2(0) = \Omega_2, \ldots$. From now on, we will use Ω as a shorthand for Ω_1. $\varepsilon_0 \notin C_0(\Omega)$, so $\psi_0(\Omega) = \varepsilon_0$. But since $\Omega \in C_0(\Omega) \subseteq C_0(\Omega+1)$, we **can** use $\psi_0(\Omega)$ in $C_0(\Omega+1)$, i.e., $\varepsilon_0 \in C_0(\Omega+1)$, so we have:

$$\begin{aligned}
\psi_0(\Omega) &= \varepsilon_0 \\
\psi_0(\Omega+1) &= \varepsilon_0 \cdot \omega = \omega^{\varepsilon_0+1} \\
\psi_0(\Omega+2) &= \varepsilon_0 \cdot \omega^2 = \omega^{\varepsilon_0+2} \\
\psi_0(\Omega+\omega) &= \varepsilon_0 \cdot \omega^\omega = \omega^{\varepsilon_0+\omega} \\
\psi_0(\Omega+\varepsilon_0) &= \varepsilon_0 \cdot \omega^{\varepsilon_0} = \varepsilon_0^2 = \omega^{\varepsilon_0+\varepsilon_0}
\end{aligned}$$

We can actually prove that $\psi_0(\Omega+\alpha) = \omega^{\varepsilon_0+\alpha}$:

Proof

We use transfinite induction:

1. Base case: $\psi_0(\Omega+0) = \omega^{\varepsilon_0+0} = \varepsilon_0$ and $0 \in C_0(\Omega+0)$

2. Successor ordinal case: If $\psi_0(\Omega+\alpha) = \omega^{\varepsilon_0+\alpha}$ and $\alpha \in C_0(\Omega+\alpha)$, then

$$\begin{aligned}\psi_0(\Omega+\alpha+1) &= \min\{\gamma \in P \mid \psi_0(\alpha) < \gamma\} \\ &= \min\left\{\omega^\xi \mid \omega^{\varepsilon_0+\alpha} < \omega^\xi\right\} \\ &= \omega^{\varepsilon_0+\alpha+1}\end{aligned}$$

   Since $\alpha, 1 \in C_0(\alpha) \subseteq C_0(\alpha+1)$, $\alpha+1 \in C_0(\alpha+1)$
3. Limit ordinal case: If for a limit ordinal λ, for all ordinals $\alpha < \lambda$, $\psi_0(\Omega+\alpha) = \omega^{\varepsilon_0+\alpha}$ and $\alpha \in C_0(\Omega+\alpha)$,

$$\begin{aligned}\psi_0(\Omega+\lambda) &= \overset{+}{\sup}\{\psi_0(\Omega+\alpha) \mid \alpha < \lambda \text{ and } \alpha \in C_0(\alpha)\} \\ &= \overset{+}{\sup}\{\omega^{\varepsilon_0+\alpha} \mid \alpha < \lambda\} \\ &= \omega^{\varepsilon_0+\lambda}\end{aligned}$$

   However, we need to show that $\Omega+\lambda \in C_0(\Omega+\lambda)$ in order to *continue* the induction to higher ordinals. If $\Omega+\lambda \in C_0(\Omega+\lambda)$, then $\lambda \in C_0(\Omega+\lambda)$ and therefore we must have $\lambda < \psi_0(\Omega+\lambda)$.

   Since $\psi_0(\Omega+\lambda) = \omega^{\varepsilon_0+\lambda}$, $\lambda < \omega^{\varepsilon_0+\lambda}$. The first ordinal that violates this property is the *second* fixed point of $\omega^\bullet = \bullet$, i.e., ε_1. Though note that for ε_1, $\psi_0(\Omega+\varepsilon_1) = \omega^{\varepsilon_0+\varepsilon_1} = \varepsilon_1$ still holds, just that $\varepsilon_1 \notin C_0(\Omega+\varepsilon_1)$

So therefore, for all $\alpha \leq \varepsilon_1$, $\psi_0(\Omega+\alpha) = \omega^{\varepsilon_0+\alpha}$

We see that our function gets "stuck" once again on $\psi_0(\Omega+\varepsilon_1)$. It only becomes "unstuck" once again when we have an ordinal $\alpha > \Omega+\varepsilon_1$ such that $\alpha \in C_0(\alpha)$. The next such ordinal is $\Omega+\Omega$, or $\Omega \cdot 2$, and we have $\psi_0(\Omega \cdot 2) = \varepsilon_1$.

We can similarly have

$$\begin{aligned}\psi_0(\Omega \cdot 3) &= \varepsilon_2 \\ \psi_0(\Omega \cdot \omega) &= \varepsilon_\omega \\ \psi_0(\Omega \cdot \varepsilon_0) &= \varepsilon_{\varepsilon_0} \\ \psi_0(\Omega \cdot \Omega) &= \zeta_0\end{aligned}$$

, replacing each "stuck" term with Ω, and in general, we can relate Buchholz's function to the Veblen function as such:

Proposition 2

$$\psi_0(\Omega^\alpha \cdot (1+\beta)) = \varphi(\alpha, \beta)$$

Proof

[WIP] probably shift to some appendix cause its quite long I think.

We can generalize this to the extended veblen function as such:

Proposition 3

$$\psi_0\left(\Omega^{(\Omega^n\alpha_n+\ldots+\Omega\alpha_1+\alpha_0)}(1+\beta)\right) = \varphi(\alpha_n, \ldots, \alpha_1, \alpha_0, \beta)$$

You can verify that these expressions containing Ω satisfies $\alpha \in C_0(\alpha)$ by expressing them in terms of ψ_1:

$$\begin{aligned}
\psi_1(0) &= \Omega \\
\psi_1(\alpha) &= \omega^{\Omega+\alpha} = \Omega \cdot \omega^\alpha \\
\psi_1(\psi_1(0)) &= \psi_1(\Omega) = \omega^{\Omega+\Omega} = \Omega^2 \\
\psi_1(\psi_1(\alpha)) &= \psi_1(\Omega \cdot \omega^\alpha) \\
\psi_1(\Omega \cdot \alpha) &= \omega^{\Omega+\Omega\cdot\alpha} = \omega^{\Omega\cdot\alpha} = \Omega^\alpha \\
\psi_1(\Omega^2) &= \Omega^\Omega
\end{aligned}$$

So for example:

$$\begin{aligned}
\zeta_0 &= \psi_0(\Omega^2) \\
&= \psi_0(\psi_1(\Omega)) \\
&= \psi_0(\psi_1(\psi_1(0)))
\end{aligned}$$

The limit of the 2-argument veblen function is Γ_0, which can be written as:

$$\begin{aligned}
\Gamma_0 &= \psi_0(\Omega^\Omega) \\
&= \psi_0(\psi_1(\Omega^2)) \\
&= \psi_0(\psi_1(\psi_1(\Omega))) \\
&= \psi_0(\psi_1(\psi_1(\psi_1(0))))
\end{aligned}$$

Continuing, we have

$$\text{SVO} = \psi_0(\Omega^{\Omega^\omega}), \text{LVO} = \psi_0\left(\Omega^{\Omega^\Omega}\right)$$

Now that we have "caught up" with the veblen function, we can go even further, with $\psi_0\left(\Omega^{\Omega^{\Omega^\Omega}}\right)$. Eventually we reach $\psi_0(\Omega^{\Omega^{\cdot^{\cdot^{\cdot}}}}) = \psi_0(\varepsilon_{\Omega+1})$, the **Bachmann-Howard Ordinal (BHO)**. (The expression $\varepsilon_{\Omega+1}$ really just means the supremum of $\left\{\Omega, \Omega^\Omega, \Omega^{\Omega^\Omega}, \ldots\right\}$).

So far we've been using ψ_1 to construct all our terms containing Ω. However, just like how we previously got "stuck" at $\psi_0(\varepsilon_0) = \varepsilon_0$, we run into the issue where $\psi_1(\varepsilon_{\Omega+1}) = \varepsilon_{\Omega+1}$:

$$\begin{aligned}
\psi_1(\varepsilon_{\Omega+1}) &= \overset{+}{\sup}\left(\psi_1(\Omega), \psi_1(\Omega^\Omega), \psi_1\left(\Omega^{\Omega^\Omega}\right), \ldots\right) \\
&= \overset{+}{\sup}\left\{\omega^{\Omega+\Omega}, \omega^{\Omega+\Omega^\Omega}, \omega^{\Omega+\Omega^{\Omega^\Omega}}, \ldots\right\} \\
&= \overset{+}{\sup}\left\{\Omega^2, \Omega^\Omega, \Omega^{\Omega^\Omega}, \ldots\right\} \\
&= \varepsilon_{\Omega+1}
\end{aligned}$$

Since $\varepsilon_{\Omega+1} \notin C_1(\varepsilon_{\Omega+1})$ and $C_0(\varepsilon_{\Omega+1}) \subset C_1(\varepsilon_{\Omega+1})$, $\varepsilon_{\Omega+1} \notin C_0(\varepsilon_{\Omega+1})$. To go further than this, we will need to invoke higher Ωs, like Ω_2, Ω_3, and so on. The next ordinal larger than $\varepsilon_{\Omega+1}$ that we can access is $\psi_2(0) = \Omega_2$, so we have $\psi_1(\Omega_2) = \varepsilon_{\Omega+1}$, and can re-express the Bachmann-Howard Ordinal as $\psi_0(\Omega_2)$ instead.

But wait, if we have $\mathrm{BHO} = \psi_0(\Omega_2)$ and $\mathrm{BHO} = \psi_0(\varepsilon_{\Omega+1}) = \psi_0(\psi_1(\Omega_2))$, which is the standard way to write it? As we have seen earlier, having the property $\alpha \in C_\nu(\alpha)$ saves us a lot of headaches by avoiding places where the function "gets stuck". So $\psi_0(\Omega_2)$ is "nicer" as $\Omega_2 \in C_0(\Omega_2)$, whereas $\psi_0(\varepsilon_{\Omega+1}) = \psi_0(\psi_1(\Omega_2))$ is "less nice" as $\varepsilon_{\Omega+1} \notin C_0(\varepsilon_{\Omega+1})$. We'll elaborate more on this when we go over the Normal Form for Buchholz's ψ. From now on, in this section, we'll color terms where $\alpha \notin C_\nu(\alpha)$ and the values that they get "stuck" at in purple.

We then have:

$$\begin{aligned}
\psi_0(\Omega_2 + 1) &= \mathrm{BHO} \cdot \omega \\
\psi_0(\Omega_2 + \omega) &= \mathrm{BHO} \cdot \omega^\omega \\
\psi_0(\Omega_2 + \mathrm{BHO}) &= \mathrm{BHO} \cdot \omega^{\mathrm{BHO}} = \mathrm{BHO}^2 \\
\psi_0(\Omega_2 + \mathrm{BHO}^2) &= \mathrm{BHO} \cdot \omega^{\mathrm{BHO}^2} = \mathrm{BHO}^{\mathrm{BHO}} \\
\psi_0(\Omega_2 + \mathrm{BHO}^{\mathrm{BHO}}) &= \mathrm{BHO} \cdot \omega^{\mathrm{BHO}^{\mathrm{BHO}}} = \mathrm{BHO}^{\mathrm{BHO}^{\mathrm{BHO}}} \\
\psi_0(\Omega_2 + \varepsilon_{\mathrm{BHO}+1}) &= \varepsilon_{\mathrm{BHO}+1}
\end{aligned}$$

Just like how $\psi_0(\Omega + \varepsilon_1) = \varepsilon_1$, we have $\psi_0(\Omega_2 + \varepsilon_{\mathrm{BHO}+1}) = \varepsilon_{\mathrm{BHO}+1}$, so we replace it with an Ω and get $\psi_0(\Omega_2 + \Omega) = \varepsilon_{\mathrm{BHO}+1}$. With this we can find the next "fixed point":

$$\begin{aligned}
\psi_0(\Omega_2 + \Omega) &= \varepsilon_{\mathrm{BHO}+1} \\
\psi_0(\Omega_2 + \Omega + \varepsilon_{\mathrm{BHO}+1}) &= \varepsilon_{\mathrm{BHO}+1}^2 \\
\psi_0(\Omega_2 + \Omega + \varepsilon_{\mathrm{BHO}+1}^2) &= \varepsilon_{\mathrm{BHO}+1}^{\varepsilon_{\mathrm{BHO}+1}} \\
\psi_0\left(\Omega_2 + \Omega + \varepsilon_{\mathrm{BHO}+1}^{\varepsilon_{\mathrm{BHO}+1}}\right) &= \varepsilon_{\mathrm{BHO}+1}^{\varepsilon_{\mathrm{BHO}+1}^{\varepsilon_{\mathrm{BHO}+1}}} \\
\psi_0(\Omega_2 + \Omega + \varepsilon_{\mathrm{BHO}+2}) &= \varepsilon_{\mathrm{BHO}+2} \\
\psi_0(\Omega_2 + \Omega + \Omega) &= \varepsilon_{\mathrm{BHO}+2}
\end{aligned}$$

This is starting to look like the veblen correspondence all over again, where

$$\psi_0(\Omega_2 + \Omega^\alpha \cdot \beta) = \varphi(\alpha, \mathrm{BHO} + \beta)$$

Now let's look at ψ_1:

$$\begin{aligned}
\psi_1(\Omega_2) &= \varepsilon_{\Omega+1} \\
\psi_1(\Omega_2 + 1) &= \varepsilon_{\Omega+1} \cdot \omega \\
\psi_1(\Omega_2 + \varepsilon_{\Omega+1}) &= \varepsilon_{\Omega+1} \cdot \omega^{\varepsilon_{\Omega+1}} = \varepsilon_{\Omega+1}^2 \\
\psi_1(\Omega_2 + \varepsilon_{\Omega+1}^2) &= \varepsilon_{\Omega+1}^{\varepsilon_{\Omega+1}} \\
\psi_1\left(\Omega_2 + \varepsilon_{\Omega+1}^{\varepsilon_{\Omega+1}}\right) &= \varepsilon_{\Omega+1}^{\varepsilon_{\Omega+1}^{\varepsilon_{\Omega+1}}} \\
\psi_1(\Omega_2 + \varepsilon_{\Omega+2}) &= \varepsilon_{\Omega+2}
\end{aligned}$$

So we have to yet again find the next ordinal $\alpha > \Omega_2 + \varepsilon_{\Omega+2}$ such that $\alpha \in C_1(\alpha)$. The first ordinal to satisfy that criteria is $\alpha = \Omega_2 \cdot 2$, so we get $\psi_1(\Omega_2 \cdot 2) = \varepsilon_{\Omega+2}$. We again have a similar correspondence going on:

$$\psi_1(\Omega_2^\alpha \cdot \beta) = \varphi(\alpha, \Omega + \beta)$$

But we don't need the veblen function anymore. The limit of what we can do with Ω_2 is:

$$\psi_1\left(\Omega_2^{\Omega_2^{\cdot^{\cdot^{\cdot}}}}\right) = \psi_1\left(\varepsilon_{\Omega_2+1}\right)$$

This is analogous the BHO earlier, where instead of $\psi_1(\varepsilon_{\Omega+1}) = \varepsilon_{\Omega+1}$, we have $\psi_2\left(\varepsilon_{\Omega_2+1}\right) = \varepsilon_{\Omega_2+1}$ (Convince yourself that this is true). So we get $\psi_1(\Omega_3) = \varepsilon_{\Omega_2+1}$, and $\psi_0\left(\varepsilon_{\Omega_2+1}\right) = \psi_0(\Omega_3)$.

We can keep going $\psi_0(\Omega_4), \psi_0(\Omega_5), \ldots$ up to $\psi_0(\Omega_\omega)$, which is known as **Buchholz's Ordinal (BO)**. This is the limit of all $\psi_0(\Omega_n)$ for finite n, and while we haven't gotten to the ordinal notation associated with Buchholz's ψ yet, there are many ordinal notations that correspond to ordinals below $\psi_0(\Omega_\omega)$.

Of course, the true limit of Buchhol'z OCF is $\psi_0\left(\Omega_\omega^{\Omega_\omega^{\cdot^{\cdot^{\cdot}}}}\right) = \psi_0\left(\varepsilon_{\Omega_\omega+1}\right)$, the **Takeuti–Feferman–Buchholz ordinal (TFBO)**, but it is not as commonly used.

3 Normal Form and Fundamental Sequences

Theorem 1

Every ordinal $\alpha < \psi_0\left(\varepsilon_{\Omega_\omega+1}\right)$ can be uniquely expressed in the form:

$$\alpha = \psi_{k_1}(\alpha_1) + \ldots + \psi_{k_n}(\alpha_n)$$

Where:
- $\alpha_1, \ldots, \alpha_n \in C_0\left(\varepsilon_{\Omega_\omega+1}\right)$
- $k_1, \ldots, k_n \leq \omega$
- $\psi_{k_1}(\alpha_1) \geq \ldots \geq \psi_{k_n}(\alpha_n)$
- $\alpha_1 \in C_{k_1}(\alpha_1), \ldots, \alpha_n \in C_{k_n}(\alpha_n)$

Like in Cantor Normal Form, α_i does not necessarily have to be in normal form.

Example 3

Which of the following are in normal form:
1. $\psi_0(0)$
2. $\psi_0(\Omega)$
3. $\psi_0(\psi_0(\Omega))$
4. $\psi_0(\psi_1(\psi_2(\Omega_3)))$

Answers:
1. $0 \in C_0(0)$, normal form
2. $\Omega = \psi_1(0) \in C_0(\Omega)$, normal form
3. $\psi_0(\Omega) = \varepsilon_0 \notin C_0(\varepsilon_0)$, not normal form Note that just because an ordinal α is expressed in normal form, that does not mean $\psi_\nu(\alpha)$ is in normal form. ($\psi_0(\Omega)$ is normal form but $\psi_0(\psi_0(\Omega))$ isn't)
4. Firstly we have $\psi_2(\Omega_3) = \varepsilon_{\Omega_2+1}$. Then we have $\psi_1(\psi_2(\Omega_3)) = \psi_1\left(\varepsilon_{\Omega_2+1}\right)$, which in normal form is $\psi_1(\Omega_3)$. We then construct the following inequality to solve for $\psi_0(\psi_1(\Omega_3))$:

$$\psi_1(\Omega_2) \leq \psi_1(\Omega_3) < \Omega_2$$
$$\psi_0(\psi_1(\Omega_2)) \leq \psi_0(\psi_1(\Omega_3)) \leq \psi_0(\Omega_2)$$
$$\text{BHO} \leq \psi_0(\psi_1(\Omega_3)) \leq \text{BHO}$$

Therefore $\psi_0(\psi_1(\Omega_3)) = \text{BHO}$. However the BHO has only one unique normal form expression: $\psi_0(\Omega_2)$. So $\psi_0(\psi_1(\psi_2(\Omega_3))) = \psi_0(\psi_1(\Omega_3))$ are both not normal form.

Before we go into the fundamental sequences of ψ, we first introduce a concept known as **cofinality**:

Definition 4

The **cofinality** of an ordinal α, denoted cof α, is the **least cardinal number** κ such that there exists a subset $S \subseteq \alpha$ of cardinality κ where α is smallest ordinal strictly greater than every element in S.

Example 4

Find the cofinality of
1. $\omega 2$
2. $\omega 2 + 3$
3. $\omega_1 = \Omega$

Answers:
1. Let $S = \{\omega + n \mid n \in \mathbb{N}\} \subset \omega 2$. S has a cardinality of ω.

   However, we still need to rule out the possibility that there might be a subset S of finite cardinality (i.e. cardinality $< \omega$) that still satisfies the criteria that $\omega 2$ is the smallest ordinal greater than every element in S

   In a subset S which has a nonzero finite cardinality, there is a largest element β, and its successor $\beta + 1$ is the smallest ordinal greater than all elements of the subset S while being less than $\omega 2$. Therefore S can't have nonzero finite cardinality.

   If S has a cardinality of 0 (i.e. $S = \emptyset = \{\}$), then 0 is the smallest ordinal greater than all elements in S. Therefore S can't have a cardinality of 0.

   Therefore the smallest cardinality of S where $\omega 2$ is the least ordinal greater than all elements in S is ω.
2. $S = \{\omega 2 + 2\} \subset \omega 2 + 3$, and S has a cardinality of 1. S can't have a cardinality of 0 since the smallest ordinal larger than nothing is 0.
3. ω_1 (and its associated cardinal $\aleph_1$) is defined as the set of all ordinals whose cardinality is $\leq \aleph_0$. So the smallest subset where ω_1 is the

Now onto fundamental sequences. A fundamental sequence for an ordinal number α with cofinality $\text{cof}(\alpha) = \beta$ is a strictly increasing sequence $(\alpha[0], \alpha[1], \ldots)$ with length β and limit α (i.e. $\alpha = \overset{+}{\sup}\{\alpha[\eta] \mid \eta < \beta\}$)

Definition 5 (**System of fundamental sequences for ψ**)

For ordinals $\alpha \leq \psi_0\left(\varepsilon_{\Omega_\omega+1}\right)$ (expressed in normal form), the ηth term of the fundamental sequence of α, $\alpha[\eta]$ is defined as such:
1. If $\alpha = \psi_{k_1}(\alpha_1) + \ldots + \psi_{k_n}(\alpha_n)$, then $\text{cof}(\alpha) = \text{cof}\left(\psi_{k_n}(\alpha_n)\right)$, and $\alpha[\eta] = \psi_{k_1}(\alpha_1) + \ldots + \left(\psi_{k_n}(\alpha_n)[\eta]\right)$
2. If $\alpha = \psi_0(0) = 1$ then $\text{cof}(\alpha) = 1$. We need a sequence of length 1, so we set $\alpha[0] = 0$

3. If $\alpha = \psi_{\nu+1}(0) = \Omega_{\nu+1}$, then $\mathrm{cof}(\alpha) = \Omega_{\nu+1}$. We need a sequence of length $\Omega_{\nu+1}$, so we set $\alpha[\eta] = \Omega_{\nu+1}[\eta] = \eta$
4. If $\alpha = \psi_\omega(0)$, then $\mathrm{cof}(\alpha) = \mathrm{cof}(\omega) = \omega$ and $\alpha[\eta] = \psi_{\omega[\eta]}(0) = \Omega_{\omega[\eta]} = \Omega_\eta$
5. If $\alpha = \psi_\nu(\beta + 1)$ then $\mathrm{cof}(\alpha) = \omega$ and $\alpha[\eta] = \psi_\nu(\beta) \cdot \eta$
6. If $\alpha = \psi_\nu(\beta)$ where β is a limit ordinal,
   (a) If $\mathrm{cof}(\beta) = \omega$ or $\mathrm{cof}(\beta) \leq \Omega_\nu$, then we have $\mathrm{cof}(\alpha) = \mathrm{cof}(\beta)$, so we need a sequence of length $\mathrm{cof}(\beta)$, so $\alpha[\eta] = \psi_\nu(\beta[\eta])$
   (b) Otherwise if $\mathrm{cof}(\beta) > \Omega_\nu$ and $\mathrm{cof}(\beta) \neq \omega$ then $\mathrm{cof}(\alpha) = \omega$. This is where the definitions diverge.

   We define μ as $\mathrm{cof}(\beta) = \Omega_{\mu+1}$.

   $\alpha[n] = \psi_\nu(\beta[\gamma[n]])$, where $\gamma[0] = 0$, and $\gamma[\eta + 1] = \psi_\mu(\beta[\gamma[\eta]])$. To illustrate this nesting process:

$$\begin{aligned}
\alpha[0] &= \psi_\nu(\beta[\gamma[0]]) = \psi_\nu(\beta[0]) \\
\alpha[1] &= \psi_\nu(\beta[\gamma[1]]) = \psi_\nu\big(\beta\big[\psi_\mu(\beta[0])\big]\big) \\
\alpha[2] &= \psi_\nu(\beta[\gamma[2]]) = \psi_\nu(\beta\big[\psi_\mu\big(\beta\big[\psi_\mu(\beta[0])\big]\big)\big]\big)
\end{aligned}$$

   In the case of $\mu = \nu$ however, we can use a shortcut: $\alpha[0] = \psi_\nu(\beta[0])$, and $\alpha[\eta + 1] = \psi_\nu(\beta[\alpha[\eta]])$

   However there are many other systems out there:

   - Dennis Maksudov:

     Same definition, except $\gamma[0] = \Omega_\mu$.

   - David_Exmachina:

     $\alpha[0] = 0$, and $\gamma[1] = 0$, otherwise $\gamma[\eta + 1]$ unchanged.

   - Buchholz's Original Paper

     Then $\alpha[\eta] = \psi_\nu\big(\beta\big[\psi_\mu(\beta[1])\big]\big)$.

     Note that this is not as commonly used, as this "fundamental sequence" is independent of η (i.e. $\alpha[0] = \alpha[1] = \ldots$).

There's a lot to go through here, so let's go through some examples:

Example 5

Calculate the fundamental sequences of the following
1. $\psi_1(0) = \Omega$
2. $\psi_0(\omega) = \omega^\omega$
3. $\psi_0(\Omega) = \varepsilon_0$
4. $\psi_0(\Omega + 1) = \varepsilon_0 \cdot \omega = \omega^{\varepsilon_0+1}$
5. $\psi_0(\Omega 2) = \varepsilon_1$
6. $\psi_0(\Omega^2) = \zeta_0$
7. $\psi_0(\Omega_2) = \mathrm{BHO}$

Before you peek into the answers, do try your hand at them!

Answers:

1. For $\psi_1(0) = \Omega$, this falls under Rule 3, $\mathrm{cof}(\psi_1(0)) = \mathrm{cof}(\Omega) = \Omega$, so we need a sequence of length Ω. As such $\psi_1(0)[\eta] = \Omega[\eta] = \eta$.

   In fact, for all $\Omega_\nu, \nu \in \mathbb{N} \setminus \{0\}, \Omega_\nu[\eta] = \eta$.

2. For $\psi_0(\omega)$, $\mathrm{cof}(\omega) = \omega$ (Rule 6(a)).

   So $\mathrm{cof}(\alpha) = \mathrm{cof}(\beta)$, and $\psi_0(\omega)[n] = \psi_0(\omega[n]) = \omega^n$. Since $\psi_0(\omega) = \omega^\omega$, $\omega^\omega[n] = \omega^n$ seems reasonable.

3. For $\psi_0(\Omega)$, $\mathrm{cof}(\Omega) > \Omega_0 = 1$, and $\mathrm{cof}(\Omega) \neq \omega$ (Rule 6(b)). So $\mathrm{cof}(\alpha) = \mathrm{cof}(\omega)$, and

$$\begin{aligned}
\psi_0(\Omega)[0] &= \psi_0(\Omega[0]) \\
&= \psi_0(0) \\
&= 1 \\
\psi_0(\Omega)[1] &= \psi_0(\Omega[\psi_0(\Omega)[0]]) \\
&= \psi_0(\Omega[1]) \\
&= \psi_0(1) \\
&= \omega
\end{aligned}$$

   So we have the following fundamental sequence, which makes sense:

$$\begin{aligned}
\varepsilon_0[0] &= 0 \\
\varepsilon_0[1] &= \omega \\
\varepsilon_0[2] &= \omega^\omega \\
\varepsilon_0[3] &= \omega^{\omega^\omega} \\
&\vdots
\end{aligned}$$

4. For $\psi_0(\Omega + 1)$, this falls under Rule 5 when insides $\beta + 1$ is a successor ordinal. In this case we just have $\psi_0(\Omega + 1)[\eta] = \psi_0(\Omega) \cdot \eta$, i.e. $(\varepsilon_0 \cdot \omega)[n] = \varepsilon_0 \cdot n$, which seems reasonable.

5. For $\psi_0(\Omega 2)$, we have $\mathrm{cof}(\Omega 2) = \Omega$ which falls under Rule 6(b), and the specical shortcut where $\mu = \nu$. We first have

$$\psi_0(\Omega)[0] = \psi_0(\Omega 2[0])$$

   But what is $\Omega 2[\eta]$? $\Omega 2[\eta] = \Omega + \Omega[\eta] = \Omega + \eta$. Continuing:

$$\begin{aligned}
\psi_0(\Omega 2)[0] &= \psi_0(\Omega 2[0]) \\
&= \psi_0(\Omega + 0) \\
&= \varepsilon_0 \\
\psi_0(\Omega 2)[1] &= \psi_0(\Omega 2[\varepsilon_0]) \\
&= \psi_0(\Omega + \varepsilon_0) \\
&= \varepsilon_0 \cdot \omega^{\varepsilon_0} \\
&= \varepsilon_0^2 = \omega^{\varepsilon_0 \cdot 2} \\
\psi_0(\Omega 2)[2] &= \psi_0(\Omega 2[\varepsilon_0^2]) \\
&= \psi_0(\Omega + \omega^{\varepsilon_0 \cdot 2}) \\
&= \varepsilon_0 \cdot \omega^{\omega^{\varepsilon_0 \cdot 2}} \\
&= \varepsilon_0^{\varepsilon_0} = \omega^{\omega^{\varepsilon_0 \cdot 2}}
\end{aligned}$$

So we have the fundamental sequence of $\varepsilon_1 = \psi_0(\Omega 2)$:

$$\begin{aligned}
\varepsilon_1[0] &= \varepsilon_0 \\
\varepsilon_1[1] &= \varepsilon_0^2 = \omega^{\varepsilon_0 \cdot 2} \\
\varepsilon_1[2] &= \varepsilon_0^{\varepsilon_0} = \omega^{\omega^{\varepsilon_0 \cdot 2}} \\
\varepsilon_1[3] &= \varepsilon_0^{\varepsilon_0^{\varepsilon_0}} = \omega^{\omega^{\omega^{\varepsilon_0 \cdot 2}}} \\
&\vdots
\end{aligned}$$

Note that this is a **different** fundamental sequence than the one we defined with veblen's φ.

6. For $\psi_0(\Omega^2)$, we have $\mathrm{cof}(\Omega^2) = \Omega$ which falls under Rule 6(b), along with the special shortcut.

   So we have

$$\psi_0(\Omega)[0] = \psi_0(\Omega^2[0])$$

   But what is $\Omega^2[\eta]$? Ω^2 itself is $\psi_1(\Omega)$, and $\psi_1(\Omega)[\eta] = \psi_1(\Omega[\eta]) = \psi_1(\eta) = \omega^{\Omega+\eta} = \Omega \cdot \omega^\eta$ (by Rule 6(a)). So $\Omega^2[\eta] = \Omega \cdot \eta$. Continuing:

$$\begin{aligned}
\psi_0(\Omega)[0] &= \psi_0(\Omega^2[0]) \\
&= \psi_0(\Omega \cdot \omega^0) \\
&= \varepsilon_0 \\
\psi_0(\Omega)[1] &= \psi_0(\Omega^2[\psi_0(\Omega^2)[0]]) \\
&= \psi_0(\Omega^2[\varepsilon_0]) \\
&= \psi_0(\Omega \cdot \omega^{\varepsilon_0}) \\
&= \psi_0(\Omega \cdot \varepsilon_0) \\
&= \varepsilon_{\varepsilon_0} \text{ (Recall } \psi_0(\Omega^\alpha \cdot (1+\beta)) = \varphi(\alpha, \beta))
\end{aligned}$$

   So we have the fundamental sequence of $\zeta_0 = \psi_0(\Omega^2)$:

$$\begin{aligned}
\zeta_0[0] &= \varepsilon_0 \\
\zeta_0[1] &= \varepsilon_{\varepsilon_0} \\
\zeta_0[2] &= \varepsilon_{\varepsilon_{\varepsilon_0}} \\
&\vdots
\end{aligned}$$

   This time, this is consistent with the one for veblen.

7. For $\psi_0(\Omega_2)$, $\mathrm{cof}(\Omega_2) = \Omega_2 > \Omega_0$ and $\mathrm{cof}(\Omega_2) \neq \omega$ so once again it falls under Rule 6(b), but we no longer have the special shortcut.

   Let's first evaluate $\gamma[\eta]$. Since $\mathrm{cof}(\Omega_2) = \Omega_2$, we have $\mu = 1$:

$$
\begin{aligned}
\gamma[0] &= 0 \\
\gamma[1] &= \psi_1(\Omega_2[\gamma[0]) \\
&= \psi_1(\Omega_2[0]) \\
&= \Omega \\
\gamma[2] &= \psi_1(\Omega_2[\gamma[1]]) \\
&= \psi_1(\Omega_2[\Omega]) \\
&= \psi_1(\Omega) \\
&= \Omega^2 \\
\gamma[3] &= \psi_1(\Omega_2[\gamma[2]]) \\
&= \psi_1(\Omega^2) \\
&= \Omega^\Omega
\end{aligned}
$$

This is a familiar pattern: $0, \Omega, \Omega^2, \Omega^\Omega, \Omega^{\Omega^\Omega}, \ldots$. Since $\psi_0(\Omega_2)[n] = \psi_0(\Omega_2[\gamma[n]]) = \psi_0(\gamma[n])$, we have the following fundamental sequence:

$$
\begin{aligned}
\psi_0(\Omega_2)[0] &= 0 \\
\psi_0(\Omega_2)[1] &= \psi_0(\Omega) = \varepsilon_0 \\
\psi_0(\Omega_2)[2] &= \psi_0(\Omega^2) = \zeta_0 \\
\psi_0(\Omega_2)[3] &= \psi_0(\Omega^\Omega) = \Gamma_0 \\
\psi_0(\Omega_2)[4] &= \psi_0\left(\Omega^{\Omega^\Omega}\right) = \mathrm{LVO} \\
&\vdots
\end{aligned}
$$

4 Extended Buchholz's function

Why do we have to stop at Ω_ω? What about $\Omega_{\omega+1}$? We can extend Buchholz's function as such to allow for even larger ordinals:

Definition 6 (**Extended Buchholz's function**)

Using the same definition of Ω_ν, we define the **Extended Buchholz's function** ψ and the set $C_\nu(\alpha)$ as such:

1. $\Omega_\nu \subseteq C_\nu(\alpha)$
2. For any two ordinals $\xi, \eta \in C_\nu(\alpha)$, their sum $\xi + \eta \in C_\nu(\alpha)$
3. For any ordinal $\xi \in C_\nu(\alpha)$, as long as $\xi < \alpha$, then $\psi_\mu(\xi) \in C_\nu(\alpha)$ for all $\mu \in C_\nu(\alpha)$
4. $\psi_\nu(\alpha) = \min\{\gamma \in \mathrm{Ord} \mid \gamma \notin C_\nu(\alpha)\}$, i.e., the smallest ordinal not inside $C_\nu(\alpha)$

It's the same as the original, just that instead of being restricted to ψ_μ with $\mu \leq \omega$, we now have $\mu \in C_\nu(\alpha)$ as highlighted in red.

With this, we can go beyond, letting $\mathrm{TFBO} = \psi_0(\Omega_{\omega+1})$. We can make $\psi_0\left(\Omega_{\varepsilon_0}\right)), \psi_0\left(\Omega_{\Gamma_0}\right)$, or even $\psi_0(\Omega_\Omega)$.

To

02 | Ordinal Notations up to $\psi_0\left(\varepsilon_{\Omega_\omega+1}\right)$

We introduce a set of ordinal notations OT corresponding to Buchholz's OCF, along with a recursive way to order them $\prec$, such that $(\mathrm{OT}, \prec)$ is order isomorphic to $\left(C_0\left(\varepsilon_{\Omega_\omega+1}\right), <\right)$

Definition 7

Let T be the set of terms and PT be the set of all prinicpal terms (the terms in PT are associated to additively principal ordinals).

- $\mathbf{0} \in T$
- Given a principal term $s \in PT$ and a term $t \in T$, $s + t \in T$
- Given a term $t \in T$, $\mu \leq \omega$, $\psi_\mu(t) \in T$ and PT

Let $\prec$ be a binary relation on T.

For terms $s, t \in T$, $s \prec t$ is defined as:
- If $s = \mathbf{0}$ then $s \prec t$ if $t \neq \mathbf{0}$
- If $t = \mathbf{0}$ then $s \prec t$ is false
- If $s = a + b$ where $a \in PT$ and $b \in T \setminus \{\mathbf{0}\}$, then
  - If $t = c + d$ for some $c \in PT$ and $d \in T \setminus \{\mathbf{0}\}$
    - If $a \neq c$ then $s \prec t$ if and only if $a \prec c$
    - If $a = c$ then $s \prec t$ if and only if $b \prec d$
  - If $t \in PT$ then $s \prec t$ if and only if $a \prec t$
- If $s = \psi_\mu(a)$ for some $a \in T, \mu \leq \omega$ then:
  - If $t = c + d$ for some $c \in PT$ and $d \in T \setminus \{\mathbf{0}\}$, then $s \prec t$ if and only if $s \preceq c$
  - If $t = \psi_\nu(b)$ for some $b \in T, \nu \leq \omega$ then:
    - If $\mu \neq \nu$ then $s \prec t$ if and only if $\mu \prec \nu$
    - If $\mu = \nu$ then $s \prec t$ if and only if $a \prec b$

For terms s, t, we have a family of the binary relations $s \in \mathbf{C}_\mu(t)$ for $\mu \leq \omega$ if and only if:

- $s \prec \psi_\mu(\mathbf{0})$
- $s = a + b$ for some $a \in PT$ and $b \in T \setminus \{\mathbf{0}\}$ and $a \in \mathbf{C}_\mu(t)$ and $b \in \mathbf{C}_\mu(t)$
- $s = \psi_\nu(a)$ for some $a \in T, \nu \leq \omega$ and $a \in \mathbf{C}_\mu(t)$ and $b \prec t$

Let OT be a subset of T. OT corresponds to the set of all ordinal notations of normal form. OT is defined as:

- $\mathbf{0} \in \mathrm{OT}$
- If $a_0, \ldots, a_k \in \mathrm{OT}$ and $a_0, \ldots, a_k \in PT$, if $a_k \preceq \ldots \preceq a_0$, their sum $a_0 + \ldots + a_k \in \mathrm{OT}$
- For a term $a \in T$, $\psi_\mu(a) \in \mathrm{OT}$ if and only if $a \in \mathrm{OT}$ and $a \in \mathbf{C}_\mu(a)$

We can now define a map to associate each ordinal notation to an ordinal below $\psi_0\left(\varepsilon_{\Omega_\omega+1}\right)$:

Definition 8

The map $o : \mathrm{OT} \mapsto C_0\left(\varepsilon_{\Omega_\omega+1}\right)$ is defined recursively as such:

1. $o(\mathbf{0}) := 0$

2. $o(a_0 + \ldots + a_k) = o(a_0) + \ldots + o(a_k)$
3. $o(\psi_\nu(a)) = \psi_\nu(o(a))$

This should be fairly self-explanatory with the notation that has been chosen essentially being red versions of the usual ordinals. OT here reprsents all ordinal notations that recursively satisfy the normal form, i.e. $\psi_a(b)$ always satisfies $b \in \boldsymbol{C}_a(b)$. We call this OT the subset of **standard form** terms.

From this, Buchholz showed that $(\{a \in \mathrm{OT} \mid a \prec \psi_1(\mathbf{0})\}, \prec)$ is order isomorphic to $\left(C_0\left(\varepsilon_{\Omega_\omega+1}\right), <\right)$, that they are well-ordered (no infinite descending sequence), and that these two sets have order type $\psi_0\left(\varepsilon_{\Omega_\omega+1}\right)$. He additionally showed that for every ordinal notation $a \prec \psi_1(\mathbf{0})$, the ordinal $o(a)$ is the order type of $\{x \in \mathrm{OT} \mid x \prec a\}$ under $\prec$.

[WIP: His proofs is not that long, maybe we can break it down here? This section isn't as formal as the one on $< \varepsilon_0$]

Note that Buchholz used $D_\nu a$ instead of our red $\psi_\nu(a)$, $\#$ instead of our $+$, and he uses $G_\nu(b)$ is somewhat analogous to $C_\nu(b)$. [WIP: What is G_v in his paper?]

Now just like how the fundamental sequence of an ordinal $\alpha[n] < \alpha$, we can similarly define a fundamental sequence $a[n]$ for each ordinal notation a, where $a[n] \prec a$. This requires defining a computable version of *cofinality* (dom in Buchholz's paper) Another thing to note about the fundamental sequences is that for a term $t \in \mathrm{OT}$, all terms in its fundamental sequence $t[n] \in \mathrm{OT}$. [WIP: no proof/proof is left as an exercise]

Recall that when defining ordinal notations $< \varepsilon_0$, we build up in layers from the bottom-up with Cantor Normal Form. Here, we go for a more "top-down" approach, where every ordinal notation $a \prec \psi_\omega(\mathbf{0})$ can be expressed as taking fundamental sequences:

$$a = \psi_\omega(\mathbf{0})[n_1][n_2]\ldots[n_k]$$

For example, to get $\psi_0(\psi_1(\mathbf{0}) + \psi_1(\mathbf{0})) \sim \varepsilon_1$, (using David_Exmachina's fundamental sequences):

- $\psi_0(\psi_\omega(\mathbf{0})) \sim \psi_0(\Omega_\omega)$
  - ⋮
  - [2] $\psi_0(\psi_2(\mathbf{0})) \sim \mathrm{BHO}$
    - ⋮
    - [4] $\psi_0(\psi_1(\psi_1(\psi_1(\mathbf{0})))) \sim \psi_0(\Omega^\Omega) = \Gamma_0$
    - [3] $\psi_0(\psi_1(\psi_1(\mathbf{0}))) \sim \psi_0(\Omega^2) = \zeta_0$
      - ⋮
      - [2] $\psi_0(\psi_1(\psi_0(\psi_1(\mathbf{0})))) \sim \psi_0(\Omega \cdot \varepsilon_0) = \varepsilon_{\varepsilon_0}$
        - ⋮
        - [2] $\psi_0(\psi_1(\psi_0(\psi_0(\mathbf{0})))) \sim \psi_0(\Omega\omega^\omega) = \varepsilon_{\omega^\omega}$
        - [1] $\psi_0(\psi_1(\psi_0(\mathbf{0}))) \sim \psi_0(\Omega\omega) = \varepsilon_\omega$
          - ⋮
          - [2] $\psi_0(\psi_1(\mathbf{0}) + \psi_1(\mathbf{0})) \sim \psi_0(\Omega 2) = \varepsilon_1$
          - [1] $\psi_0(\psi_1(\mathbf{0})) \sim \psi_0(\Omega) = \varepsilon_0$
          - [0] $\psi_0(\mathbf{0}) \sim 1$
        - [0] $\psi_0(\psi_1(\mathbf{0})) \sim \psi_0(\Omega) = \varepsilon_0$
      - [1] $\psi_0(\psi_1(\mathbf{0})) \sim \psi_0(\Omega) = \varepsilon_0$
      - [0] $\mathbf{0} \sim 0$
    - [2] $\psi_0(\psi_1(\mathbf{0})) \sim \psi_0(\Omega)$

- [1] $\psi_0(\mathbf{0}) \sim 1$
- [0] $\mathbf{0} \sim 0$

- [1] $\psi_0(\psi_1(\mathbf{0})) \sim \psi_0(\Omega) = \varepsilon_0$
- [0] $\psi_0(\psi_0(\mathbf{0})) \sim \psi_0(1) = \omega$

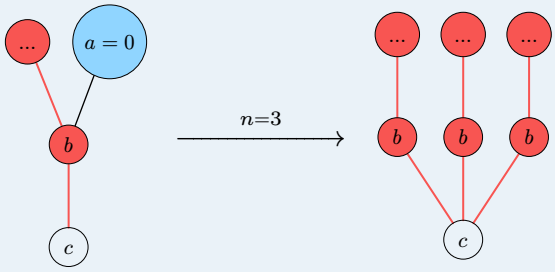
1 Buchholz's Hydra

Definition 9

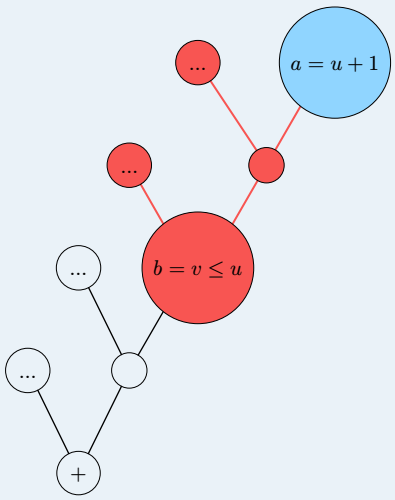
This is somewhat similar to the Kirby-Paris hydra game we have earlier, but each node is labelled with an ordinal $\leq \omega$, the child nodes of the root must be labelled 0.

At each step, we choose a leaf node a to chop off. The hydra chooses some $n \in \mathbb{N}$.

1. If a has label 0, we proceed as in Kirby-Paris' game. Call the node's parent b, and its grandparent c (if it exists). First we delete a. If c exists (i.e. b is not the root), we make n copies of b and all its children and attach them to c:

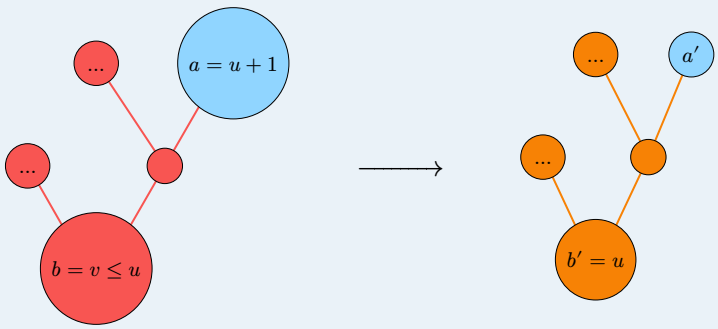


2. If a has label $u + 1$, we go down the tree looking for a node b whose label is $v \leq u$:

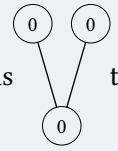
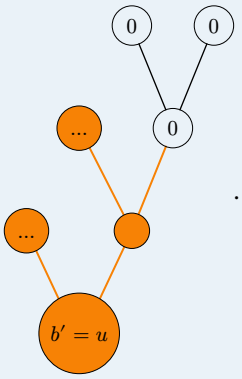


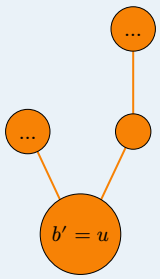
Let the subtree rooted in b be called S. The nodes highlighted in red (and node a highlighted in blue) make up S.

Let's make a copy of S called S', and re-label the b inside S' to u.



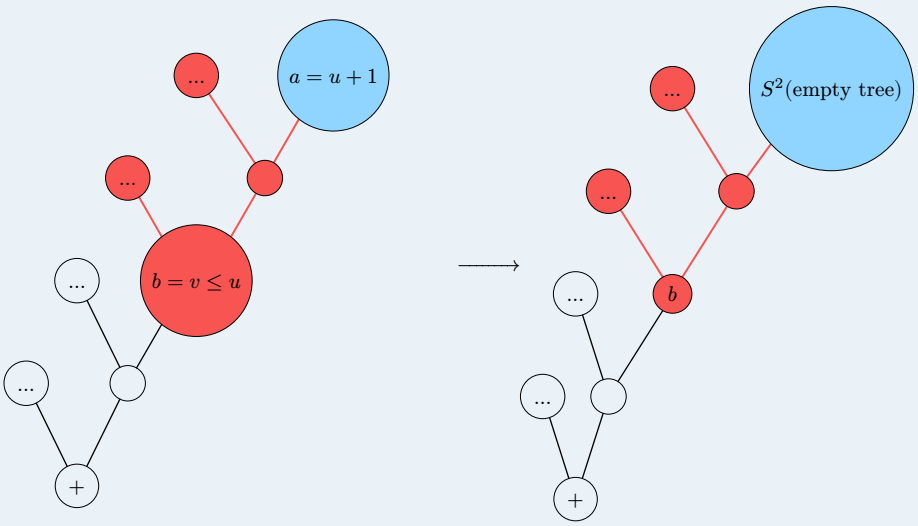
Now let's define the function $S(T)$ that takes in a tree T, and replaces a' with T

For example, if T is  then $S(T)$ is  .

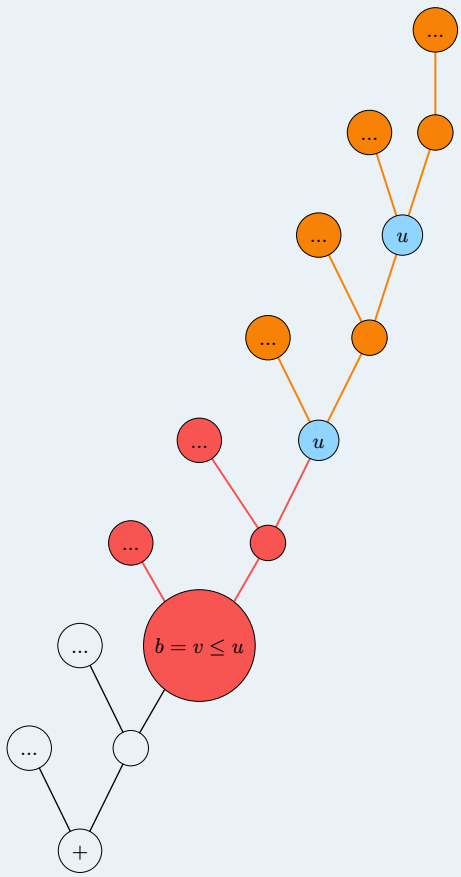
Similarly, if T is the empty tree then $S(T)$ just becomes  .

When we cut off a head of label $u + 1$, we replace it with $S^n(\text{empty tree})$ where S^n denotes function iteration.

An illustration of the whole process with $n = 2$ is as such:



Which becomes



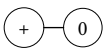
3. If a has label ω, replace the label ω with $n + 1$

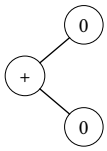
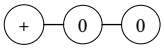
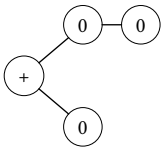
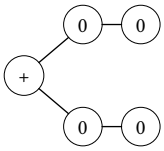
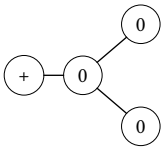
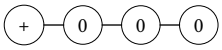
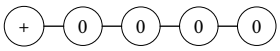
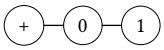
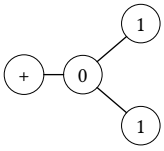
Similarly to the Kirby-Paris Hydra, we can associate each hydra with an ordinal notation term T by the following rules:

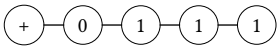
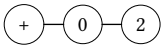
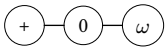
1. For each leaf node with a label u, we replace it with the ordinal notation $\psi_u(0)$
2. For a node with label u with children of ordinal notations $a_0, \ldots, a_k$, we replace it with the ordinal notation $\psi_u(a_0 + \ldots + a_k)$
3. At the root node, the resulting ordinal expressions of each of the root's children $a_0, \ldots, a_k$ are summed up: $a_0 + \ldots + a_k$

Note that case 2 in the definition above is different from the original definiton that Buchholz gave. This difference basically corresponds to the differences in the system of fundamental sequence (Specifically "Rule 6(b)").

Here are some examples of Buchholz's hydra and the associated ordinal notations and ordinals. The hydras have been rotated to save space.

Buchholz Hydra	Associated Ordinal Notation	Associated Ordinal
	$\mathbf{0}$	0
	$\psi_0(\mathbf{0})$	1

Buchholz Hydra	Associated Ordinal Notation	Associated Ordinal
	$\psi_0(\mathbf{0}) + \psi_0(\mathbf{0})$	2
	$\psi_0(\psi_0(\mathbf{0}))$	ω
	$\psi_0(\psi_0(\mathbf{0})) + \psi_0(\mathbf{0})$	$\omega + 1$
	$\psi_0(\psi_0(\mathbf{0})) + \psi_0(\psi_0(\mathbf{0}))$	$\omega 2$
	$\psi_0(\psi_0(\mathbf{0}) + \psi_0(\mathbf{0}))$	ω^2
	$\psi_0(\psi_0(\psi_0(\mathbf{0})))$	ω^ω
	$\psi_0(\psi_0(\psi_0(\psi_0(\mathbf{0}))))$	ω^{ω^ω}
	$\psi_0(\psi_1(\mathbf{0}))$	$\psi_0(\Omega) = \varepsilon_0$
	$\psi_0(\psi_1(\mathbf{0}) + \psi_1(\mathbf{0}))$	$\psi_0(\Omega 2) = \varepsilon_1$

Buchholz Hydra	Associated Ordinal Notation	Associated Ordinal
	$\psi_0(\psi_1(\psi_1(\mathbf{0})))$	$\psi_0(\Omega^2) = \zeta_0$
	$\psi_0(\psi_1(\psi_1(\psi_1(\mathbf{0}))))$	$\psi_0(\Omega^\Omega) = \Gamma_0$
	$\psi_0(\psi_1(\psi_1(\psi_1(\psi_1(\mathbf{0})))))$	$\psi_0\left(\Omega^{\Omega^\Omega}\right) = \mathrm{LVO}$
	$\psi_0(\psi_2(\mathbf{0}))$	$\psi_0(\Omega_2) = \mathrm{BHO}$
	$\psi_0(\psi_\omega(\mathbf{0}))$	$\psi_0(\Omega_\omega) = \mathrm{BO}$

By restricting Buchholz Hydra to only those corresponding to OT (rather than all of T), we can prove the termination of Buchholz Hydras.

2 The Hyper Primitive Sequence System (HPrSS)

The Hyper Primitive Sequence System (HPrSS) is an extension of the Primitive Sequence System (PrSS), and much like how PrSS can be thought of as a linearized version of the Kirby-Paris Hydra, HPrSS can be thought of as a linearized version of Buchholz's Hydra.

Definition 10

A sequence $s = (s_1, s_2, \ldots, s_m)$ is a valid HPrSS sequence (i.e. $s \in T_{\mathrm{HPrSS}}$) if s is either empty $()$ or starts with 0: $(0, \ldots)$.

We then define its fundamental sequence $s[n]$, which can be used to define a fast-growing function similarly to PrSS.

We then take the last element of the sequence, s_m, and find its **parent**, the rightmost s_i such that $s_i < s_m$ (same as PrSS). We then keep finding that parent's parent, and so on until we can no longer find a parent.

We let $m = m_0$, then denote the index of s_{m_0}'s parent as m_1, the index s_{m_1}'s parent as m_2, and so on.
We end up with a table like this:

k	m_k	s_{m_k}
0	$m_0 = m$	s_{m_0}
1	m_1	s_{m_1}
$\vdots$	$\vdots$	$\vdots$
k	m_k	s_{m_k}

Now let's define a **difference sequence** $N = (N_0, N_1, ..., N_{k-1})$, where $N_i = S_{m_i} - S_{m_{i+1}}$.

1. If $N_0 = 1$, then we proceed as normal PrSS, with **good root** being $\left(s_1, ..., s_{m_1-1}\right)$ and **bad root** being $\left(s_{m_1}, ..., s_{m_0-1}\right)$. and if $m_1 = 0$ then good root is empty.
2. If $N_0 \neq 1$, then we take the difference sequence $N = (N_0, N_1, ..., N_{k-1})$, and try to find the first element in the rest of the sequence N_i such that $N_i < N_0$. Put more formally,

   $$p = \min\{i \mid i \in N \text{ and } 0 < i \leq k-1 \text{ and } N_i < N_0\}$$

   We then define r and **ascension factor** δ as such
   1. If p doesn't exist/no such i exists (i.e., $N_i \not< N_0$ for all $0 < i \leq k-1$), then we have $r = 1$, and ascension factor $\delta = S_{m_0} - 1$
   2. If p does exist, then $r = m_p$ and ascension factor $\delta = S_{m_0} - S_{m_p} - 1$.

   We define **good root** $G = (s_1, ..., s_{r-1})$ for $r > 1$, or just empty $G = ()$ if $r = 1$. We then define **bad root** as a **function** $B(n) = \left(s_r + n \cdot \delta, ..., s_{m_0-1} + n \cdot \delta\right)$, i.e. $\left(s_r, ..., s_{m_0-1}\right)$ but add $n \cdot \delta$ to each element.

   Finally, we define $s[n]$ as the concatenation of $G, B(0), ..., B(n)$.

In the original $N_0 = 1$ case, where normal PrSS rules apply, we can think of it as $r = m_1$ and $\delta = 0$.

Just like in PrSS, we can define a fast-growing function from these fundamental sequences also labelled $s[n]$:

$$s[n] = (G, B_0, ..., B_n)[f(n)]$$

where $f(n)$ is a fast-growing function, usually $f(n) = n + 1$.

Example 6

Expand the following HPrSS sequences:
1. $(0, 2, 1)[n]$
2. $(0, 2)[n]$
3. $(0, 2, 1, 3)[n]$

Answers:
1. For $s = (0, 2, 1)$, the parent of $s_3 = 1$ is $s_1 = 0$, so we have:

$\boldsymbol{k}$	$\boldsymbol{m_k}$ (indices of parents)	$\boldsymbol{s_{m_k}}$ (values of parents)
0	$m_0 = 3$	$s_{m_0} = s_3 = 1$
1	$m_1 = 1$	$s_{m_1} = s_1 = 0$

Our difference sequence is just $N = (1)$, so by normal PrSS expansion we have good root $G = ()$ and bad root $B = (0, 2)$, and as such

$$(0, 2, 1)[n] = \underbrace{(0, 2, 0, 2, ...)}_{(0,2) \text{ repeated } n \text{ times}}$$

Now using our new system, we have $r = m_1 = 1$ so we still have empty good root $G = ()$, and an ascension factor of $\delta = 0$, so our bad root function

$$\begin{aligned} B(n) &= \left(s_r + n \cdot 0, ..., s_{m_0 - 1} + n \cdot 0\right) \\ &= \left(s_1, ..., s_{m_0 - 1}\right) \\ &= (0, 2) \end{aligned}$$

So we end up with the same result.

2. For $s = (0, 2)$, the parent of $s_2 = 2$ is $s_1 = 0$, so we have

$\boldsymbol{k}$	$\boldsymbol{m_k}$ (indices of parents)	$\boldsymbol{s_{m_k}}$ (values of parents)
0	$m_0 = 2$	$s_{m_0} = s_2 = 2$
1	$m_1 = 1$	$s_{m_1} = s_1 = 0$

Our difference sequence is $N = (2)$, so we fall under the sub-case 1, where there exists no i such that $N_i < N_0$. We then have $r = 1$ which leads to an empty good root $G = ()$, and an ascension factor of $\delta = s_{m_0} - 1 = 2 - 1 = 1$. So our bad root function is:

$$\begin{aligned} B(n) &= \left(s_r + n \cdot \delta, ..., s_{m_0 - 1} + n \cdot \delta\right) \\ &= (s_1 + n \cdot 1) \\ &= (0 + n) \\ &= (n) \end{aligned}$$

So concatenating $G, B(0), ..., B(n)$, we concatenate $(), (0), (1), ..., (n)$:

$$(0, 2)[n] = \underbrace{(0, 1, 2, 3, ...)}_{\text{of length } n}$$

3. For $s = (0, 2, 1, 3)$, the parent of $s_4 = 3$ is $s_3 = 1$, whose parent is $s_1 = 0$, so we have

k	m_k (indices of parents)	s_{m_k} (values of parents)
0	$m_0 = 4$	$s_{m_0} = s_4 = 3$
1	$m_1 = 2$	$s_{m_1} = s_2 = 1$
2	$m_2 = 1$	$s_{m_2} = s_1 = 0$

Our difference sequence is $N = (2, 1)$. Since for $i = 1$, $N_i < N_0$, we have $p = 1$ can use the second sub-case, and we have $r = m_p = m_1$ and $\delta = s_{m_0} - s_{m_p} - 1 = s_4 - s_1 - 1 = 1$.

You should be able to perform good root and bad root calculations by now, so concatenating $G = (0, 2)$ and $B(n) = (n + 1)$ we have:

$$(0, 2, 1, 3)[n] = (0, 2, 1, 2, 3, \ldots)$$

Definition 11

The **standard form** of sequences in HPrSS $\mathrm{OT}_{\mathrm{HPrSS}}$ is a subset of T_{HPrSS} defined as such;

1. For any $n \in \mathbb{N}$, $(0, n) \in \mathrm{OT}_{\mathrm{HPrSS}}$.
2. For any $S \in \mathrm{OT} \setminus \{()\}$ and $n \in \mathbb{N}$, $s[n] \in \mathrm{OT}_{\mathrm{HPrSS}}$

Again, the OT here is built in a "top-down" approach. It might not seem like a "top-down" approach, but it will make more sense when we define a correspondence with the ordinal notation.

2.1 Association with Ordinal Notation and Buchholz Hydras

Definition 12

We can define a map $\mathrm{Trans} : T_B \mapsto T_{\mathrm{HPrSS}}$, where T_B is the set of terms associated to Buchholz's OCF except those containing $\psi_\omega()$, and T_{HPrSS} is the set of all HPrSS terms.

$\mathrm{Trans}(t)$ is defined as such:

- $\mathrm{Trans}(\mathbf{0}) = ()$
- Suppose $t = t_0 + \psi_u(t_1)$ for some $u \in \mathbb{N}$, and $t_0, t_1 \in T_B$
  - If $t_1 = \psi_0(\mathbf{0}) \cdot m$ for some $m \in \mathbb{N}$, then

    $$\mathrm{Trans}(t_0 + \psi_u(\psi_0(\mathbf{0}) \cdot m)) = \mathrm{Trans}(t_0)^\frown(u, \underbrace{u + 1, \ldots, u + 1}_{m})$$

    where $^\frown$ represents concatenation. Note that if $t = \psi_u(\mathbf{0})$ this becomes $t_0 = \text{empty}, m = 0$, and as such:

    $$\mathrm{Trans}(\psi_u(0)) = (u)$$

  - Otherwise, $\mathrm{Trans}(t)$ is the concatenation of $\mathrm{Trans}(t_0)$, (u), and $\mathrm{Trans}(t_1)$ with $(u + 1)$ added to each element.

    $$\mathrm{Trans}(t_0 + \psi_u(t_1)) = \mathrm{Trans}(t_0)^\frown(u)^\frown[\mathrm{Trans}(t_1)]^{+(u+1)}$$

    (We use $[s]^{+k}$ to represent adding k to every element of a sequence s).
    Again, we can see that if $t = \psi_0(\psi_v(\mathbf{0}))$, $u = 0$ and

$$\mathrm{Trans}(\psi_0(\psi_v(\mathbf{0}))) = (0)^\frown[\mathrm{Trans}(\psi_v(\mathbf{0}))]^{+1} = (0, v+1)$$

If we restrict Trans to $\mathrm{OT}_B \to$, Trans is an *order isomorphism* from $(\mathrm{OT}_B, \prec)$ and $\left(\mathrm{OT}_{\mathrm{HPrSS}}, \underset{\mathrm{lex}}{\prec}\right)$. We can then prove the termination of all normal form HPrSS sequences by showing that $s[n] \underset{\mathrm{lex}}{\prec} s$:

Lemma 8

Let OT_B be the set of ordinal terms associated with Buchholz's OCF OT except those containing $\psi_\omega()$. If we restrict Trans's domain to OT_B, it satisfies the following:

1. It is a bijective map onto $\mathrm{OT}_{\mathrm{HPrSS}}$, the set of normal form HPrSS sequences
2. For any $t \in \mathrm{OT}_B \setminus \{\mathbf{0}\}$ and $n \in \mathbb{N}$, and there exists a $t' \in \mathrm{OT}_B$ such that $\mathrm{Trans}(t') = \mathrm{Trans}(t)[n]$ and $t' \prec t$. If $n > 0$ then such a $t[n-1] < t' \leq t[n]$

Proof

[WIP: I do not understand the proof given in the Googology wiki]. Obviously $\mathrm{Trans}(t) = \mathrm{Trans}(t'[n])$, but I don't know how to prove $\mathrm{Trans}(t'[n]) = \mathrm{Trans}(t')[n]$

As we saw earlier in PrSS, $(0, 1, 2, 3, \ldots)$ is equivalent to $\omega^{\omega^{\cdot^{\cdot^{\cdot}}}} = \varepsilon_0$. Since we now have $(0, 2)[n]$ which expands to $(0, 1, 2, 3, \ldots)$, we have $(0, 2)$ corresponding to ε_0. In fact, we can expand upon this and define larger and larger ordinals: We denote $\mathbf{1}$ as $\psi_0(\mathbf{0})$, $\mathbf{\Omega}$ as $\psi_1(\mathbf{0})$, and $\mathbf{\Omega}_v$ as $\psi_v(\mathbf{0})$.

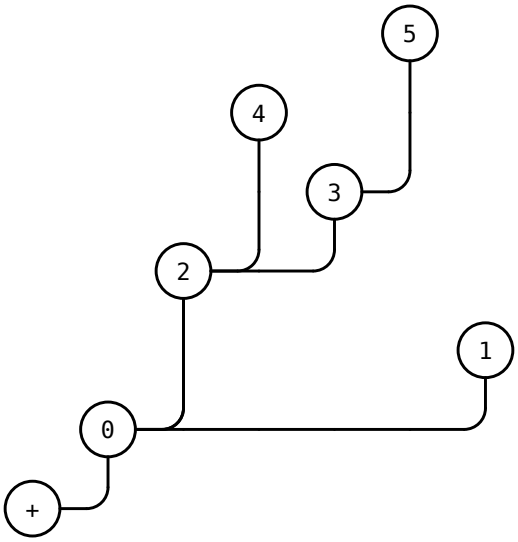
HPrSS	Ordinal Notation	Common expression
$()$	$\mathbf{0}$	0
(0)	$\psi_0(\mathbf{0})$	1
$(0, 0)$	$\psi_0(\mathbf{0}) + \psi_0(\mathbf{0})$	2
$(0, 1)$	$\psi_0(\psi_0(\mathbf{0}))$	ω
$(0, 1, 2)$	$\psi_0(\psi_0(\psi_0(\mathbf{0})))$	ω^ω
$(0, 1, 2, 3)$	$\psi_0(\psi_0(\psi_0(\psi_0(\mathbf{0}))))$	ω^{ω^ω}
$(0, 2)$	$\psi_0(\mathbf{\Omega})$	ε_0
$(0, 2, 1)$	$\psi_0(\mathbf{\Omega} + \psi_0(\mathbf{0}))$	$\varepsilon_0 \cdot \omega$
$(0, 2, 1, 2)$	$\psi_0(\mathbf{\Omega} + \psi_0(\psi_0(\mathbf{0})))$	$\varepsilon_0 \cdot \omega^\omega$
$(0, 2, 1, 2, 3)$	$\psi_0(\mathbf{\Omega} + \psi_0(\psi_0(\psi_0(\mathbf{0}))))$	$\varepsilon_0 \cdot \omega^{\omega^\omega}$
$(0, 2, 1, 3)$	$\psi_0(\mathbf{\Omega} + \psi_0(\mathbf{\Omega}))$	ε_0^2
$(0, 2, 1, 3, 2, 4)$	$\psi_0(\mathbf{\Omega} + \psi_0(\mathbf{\Omega} + \psi_0(\mathbf{\Omega})))$	$\varepsilon_0^{\varepsilon_0}$
$(0, 2, 2)$	$\psi_0(\mathbf{\Omega} + \mathbf{\Omega})$	ε_1

There is a somewhat weird way I came up with to associate HPrSS with Buchholz Hydras

Suppose we have a sequence: $(0, 2, 4, 3, 5, 1)$. We label each element's parent:

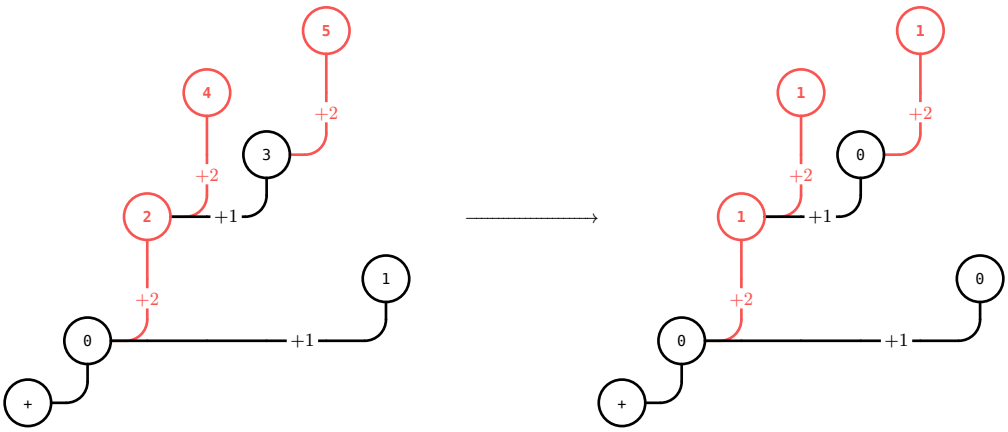
$$(0, 2, 4, 3, 5, 1)$$

Then we draw a tree representing the parent relationships:



So for example, 5's parent is 3, whose parent is 2, whose parent is 0.

Now let's consider the different in values. If the a node's parent's value is 1 less than its own value, we label the node 0. If a node's parent's value is 2 less than its own value, we label the node 1. And in general, if a node's parent's value is n less than its own value, we label the node $n - 1$. This results in the following Buchholz Hydra:



3 The Pair Sequence System (PSS)

The Pair Sequence System (PSS) can also be considered an extension to the primitive sequence system, where instead of it being a sequence of natural numbers, the Pair Sequence System can be thought og as a sequence of *pairs* of natural numbers. There are two common ways to write PSS, either inline, like $(0,0)(1,1)(2,1)(2,0)$ or as a matrix like so:

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 0 & 1 & 1 & 0 \end{pmatrix}$$

The PSS is not as comptabile with Buchholz's ordinal notation as HPrSS. Nonetheless it is possible to show a correspondence with PSS and Buchholz's ordinal notation.

divergence at $\psi_0(\Omega_\omega \cdot (\Omega + 1))$

4 Patcail's Hydra/Mini-Nuclear Array Notation

5 Ordinal notation associated with Extended Buchholz's function

Definition 13

Let T be the set of terms and PT be the set of all prinicpal terms (terms whose associated ordinal are additively principal ordinals).

- $\mathbf{0} \in T$
- Given a principal term $s \in PT$ and a term $t \in T$, $s + t \in T$
- Given terms $s, t \in T$, $\psi_s(t) \in T$ and PT

Let $\prec$ be a binary relation on T.

For terms $s, t \in T$, $s \prec t$ is defined as:
- If $s = \mathbf{0}$ then $s \prec t$ if $t \neq \mathbf{0}$
- If $t = \mathbf{0}$ then $s \prec t$ is false
- If $s = a + b$ where $a \in PT$ and $b \in T \setminus \{\mathbf{0}\}$, then
  - If $t = c + d$ for some $c \in PT$ and $d \in T \setminus \{\mathbf{0}\}$
    - If $a \neq c$ then $s \prec t$ if and only if $a \prec c$
    - If $a = c$ then $s \prec t$ if and only if $b \prec d$
  - If $t \in PT$ then $s \prec t$ if and only if $a \prec t$
- If $s = \psi_a(b)$ for some $a, b \in T$ then:
  - If $t = c + d$ for some $c \in PT$ and $d \in T \setminus \{\mathbf{0}\}$, then $s \prec t$ if and only if $s \preceq c$
  - If $t = \psi_c(d)$ for some $c, d \in T$ then:
    - If $a \neq c$ then $s \prec t$ if and only if $a \prec c$
    - If $a = c$ then $s \prec t$ if and only if $b \prec d$

For terms s, t, u, we define the ternary relation $s \in C_t(u)$ if and only if:

- $s \prec \psi_t(\mathbf{0})$
- $s = a + b$ for some $a \in PT$ and $b \in T \setminus \{\mathbf{0}\}$ and $a \in C_t(u)$ and $b \in C_t(u)$
- $s = \psi_a(b)$ for some $a, b \in T$ and $a \in C_t(u)$ and $b \in C_t(u)$ and $b \prec u$

Let OT be a subset of T. OT corresponds to the set of all ordinal notations of normal form. For a term s, $s \in \mathrm{OT}$ is defined as:

- If $s = \mathbf{0}$ then $s \in \mathrm{OT}$
- If $s = a + b$ for some $a, b \in PT$, then $s \in \mathrm{OT}$ if and only if $a, b \in \mathrm{OT}$ and $b \preceq a$
- If $s = a + b + c$ for some $a, b \in PT$ and $c \in T \setminus \{\mathbf{0}\}$ then $s \in \mathrm{OT}$ if and only if $a, (b + c) \in \mathrm{OT}$ and $b \preceq a$
- If $s = \psi_a(b)$ for some $a, b \in T$ then $s \in \mathrm{OT}$ if and only if $a, b \in \mathrm{OT}$ and $b \in C_a(b)$

03 Introduction to Bashicu matrix system

[WIP] This shit is too powerful maybe just do an introductory chapter

- ## 1 Definition

- ## 2 Wrangling with upgrading